

Presented: May 19, 2026

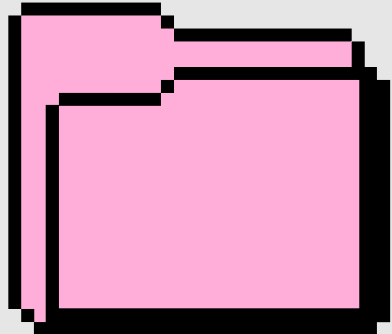
NEW YORK CITY



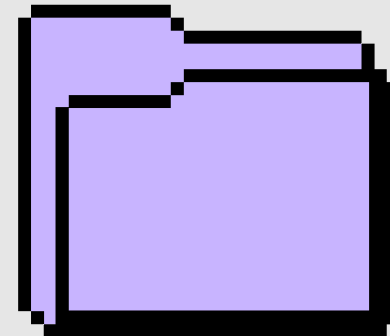
TEMPERATURE PREDICTION MODEL

By: Fatoumata Drammeh/CSC44800 Artificial Intelligence

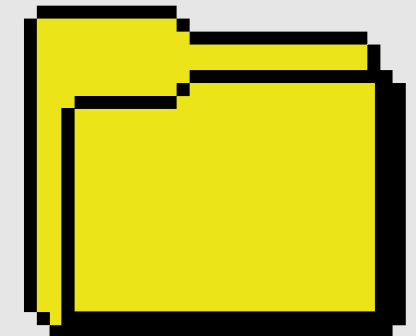
IN THIS DECK



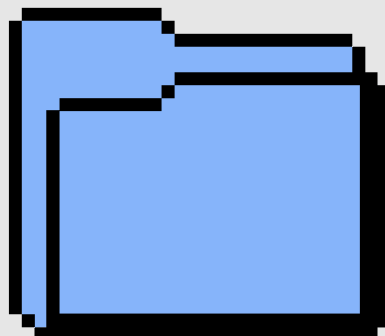
Exploratory Data
Analysis



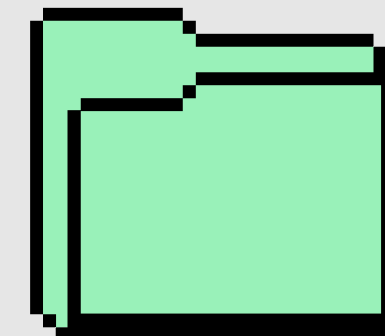
Seq2Seq LSTM Model



Comparisons



Many-to-One LSTM
Model



Prophet Model

https://www.kaggle.com/datasets/danbraswell/new-york-city-weather-18692022

DATASET

New York City Weather: A 154-Year Retrospective

27 Code

Data Card Code (6) Discussion (1) Suggestions (0)

NYC_Central_Park_weather_1869-2022.csv (1.33 MB)

Download Open in Colab

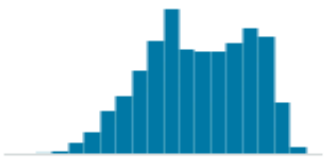
Detail Compact Column

6 of 6 columns

About this file

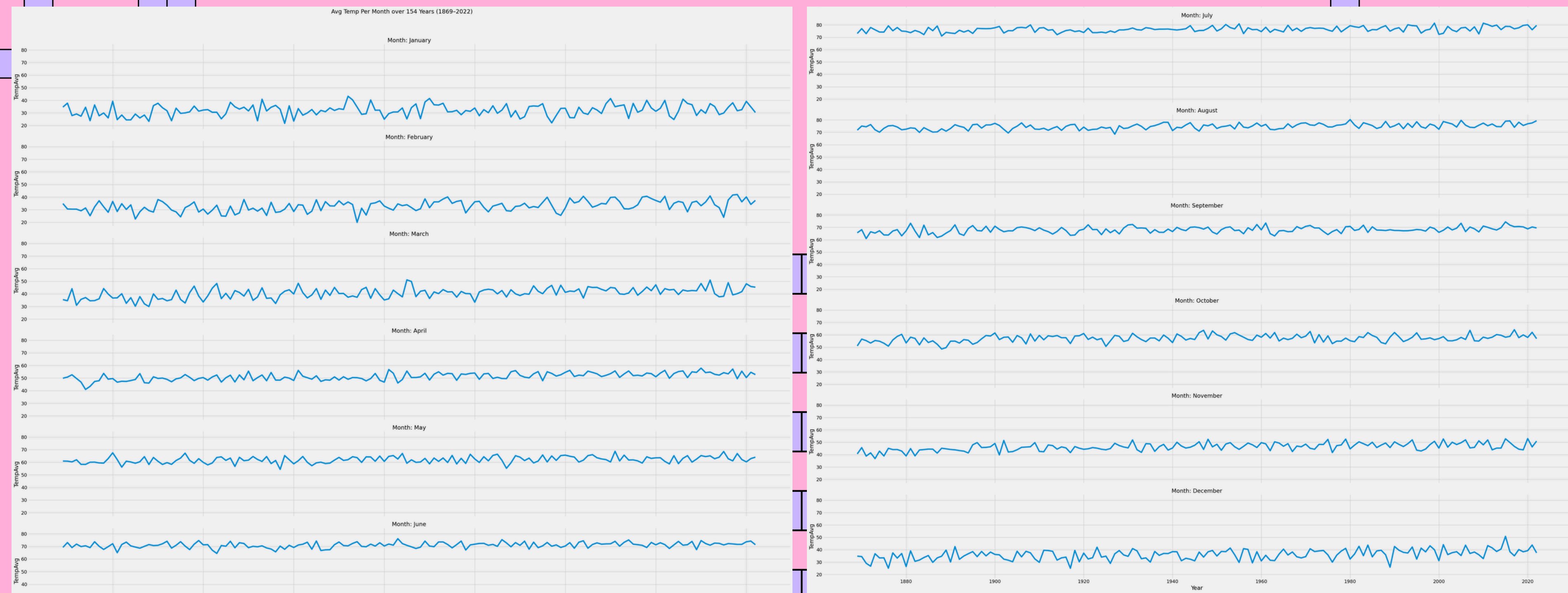
Suggest Edits

This dataset represents a 154 year record of continuous weather observations from Central Park in NYC.

DATE	# PRCP	# SNOW	SNWD	# TMIN	# TMAX
1869-01-01 to 2022-12-29	Precipitation (inches)	Snowfall (inches)	Snow depth (inches) missing before 1912	Minimum temperature (deg F)	Maximum (deg F)
			0 66% [null] 29% Other (2562) 5%		
1868-12-31 2022-12-28	0 8.28	0 27.3		-15 87	2
1869-01-01	0.75	9		19	29
1869-01-02	0.03	0		21	27
1869-01-03	0	0		27	35
1869-01-04	0.18	0		34	37
1869-01-05	0.05	0		37	43
1869-01-06	0	0		34	38
1869-01-07	0	0		35	48
1869-01-08	0	0		40	54
1869-01-09	0	0		38	48

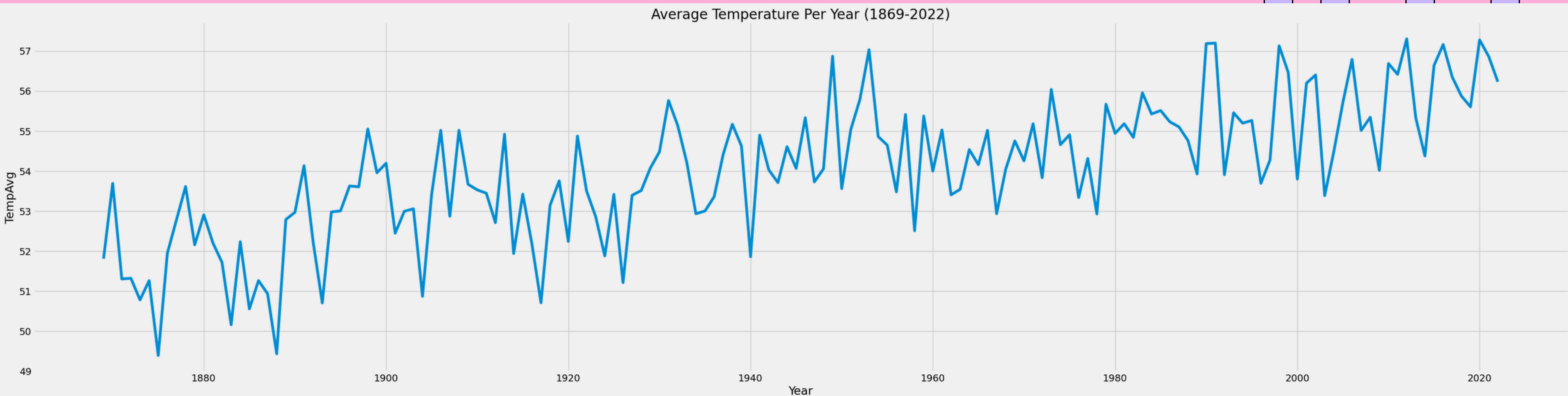
WHAT IS THE CURRENT SITUATION?

Average Temperature Per Month over 154 Years(1869-2022)



WHAT IS THE CURRENT SITUATION?

Average Temperature Per Year (1869–2022)



MANY-TO-ONE LSTM MODEL

[[1] [2] [3] [4] [5]] → [6]

[[2] [3] [4] [5] [6]] → [7]

[[3] [4] [5] [6] [7]] → [8]

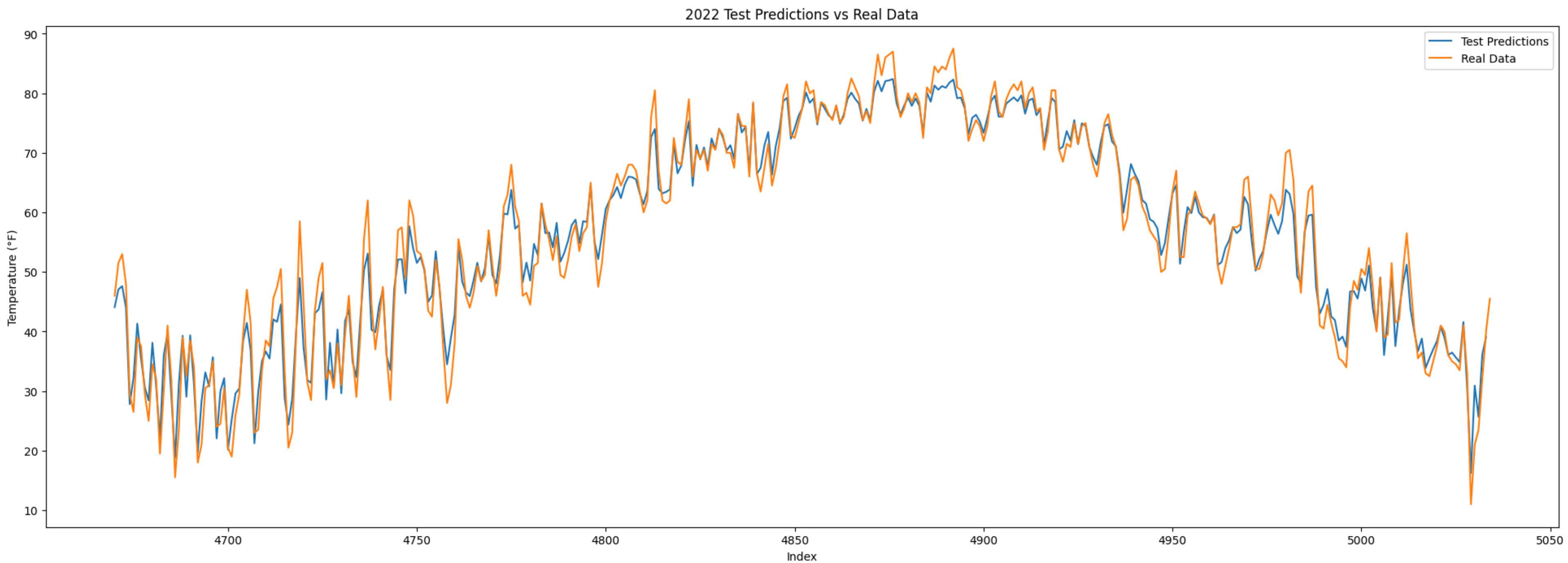
[[4] [5] [6] [7] [8]] → [9]

MANY-TO-ONE LSTM MODEL

Model #	# of Epochs	Window Size	Val Loss	RMSE
1	10	7	28.7086	5.5745
2	10	30	28.3182	5.5243
3	10	60	27.4822	5.4437
4	10	90	27.3018	5.4231
5	20	60	27.547	5.4186
6	20	90	27.1158	5.9389
7	30	60	27.3613	5.4036

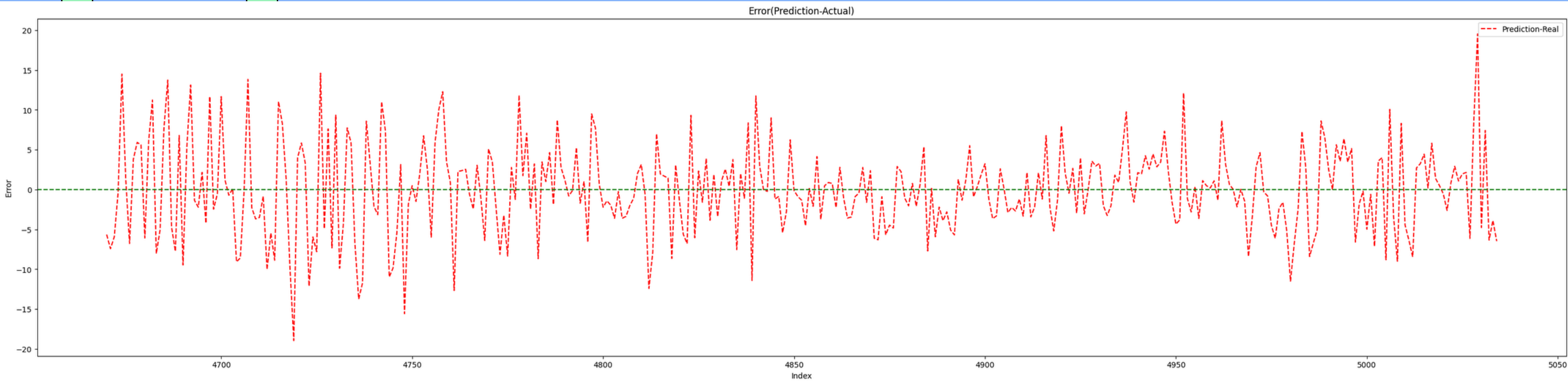
MANY-TO-ONE LSTM MODEL

2022 Real Temperatures vs Predictions



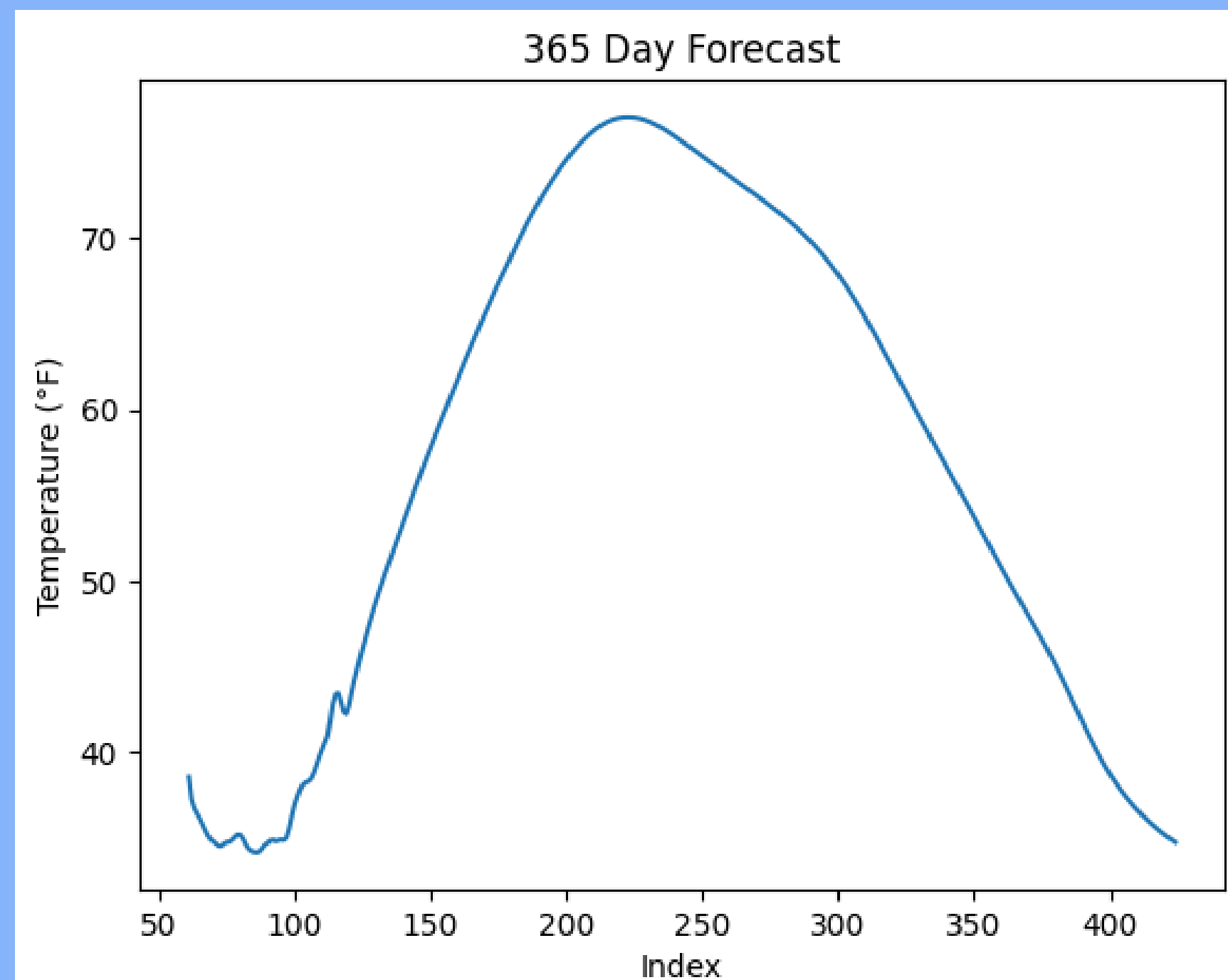
MANY-TO-ONE LSTM MODEL

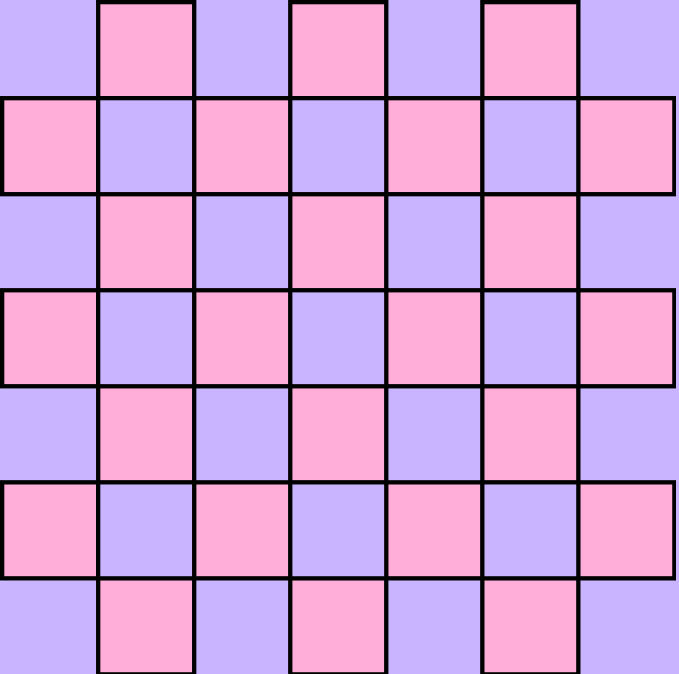
2022 Error(Prediction-Actual)



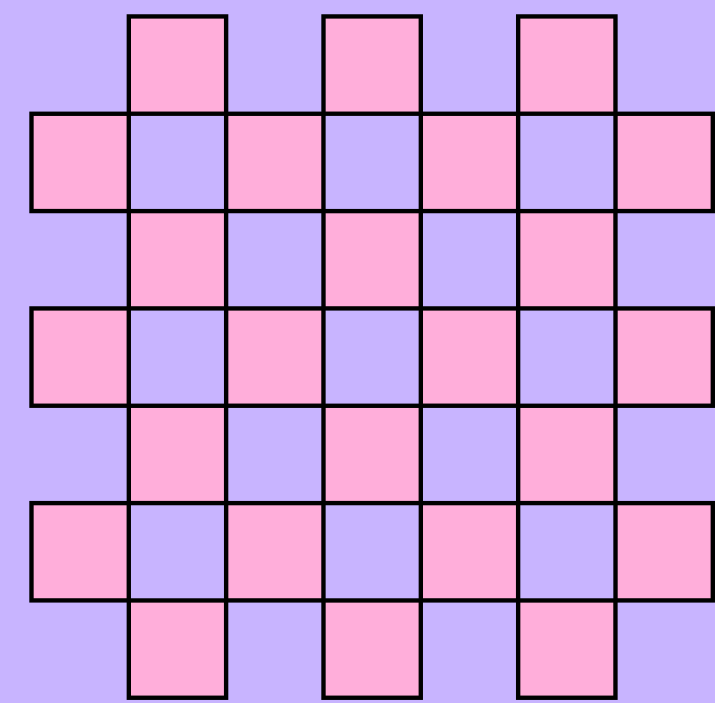
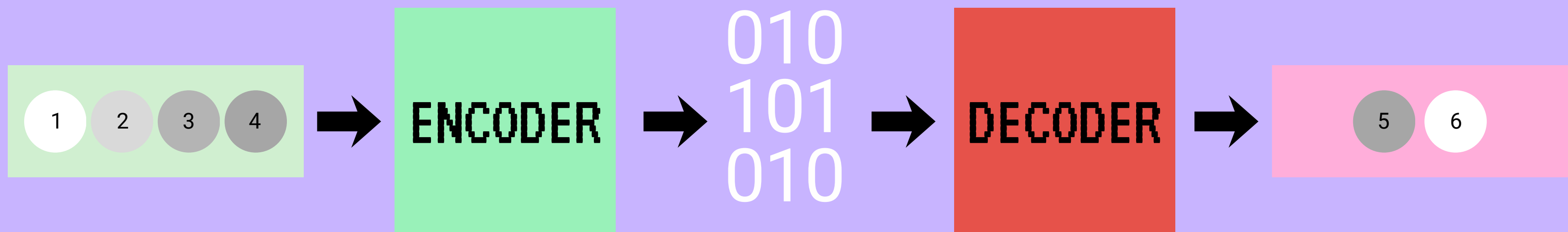
MANY-TO-ONE LSTM MODEL

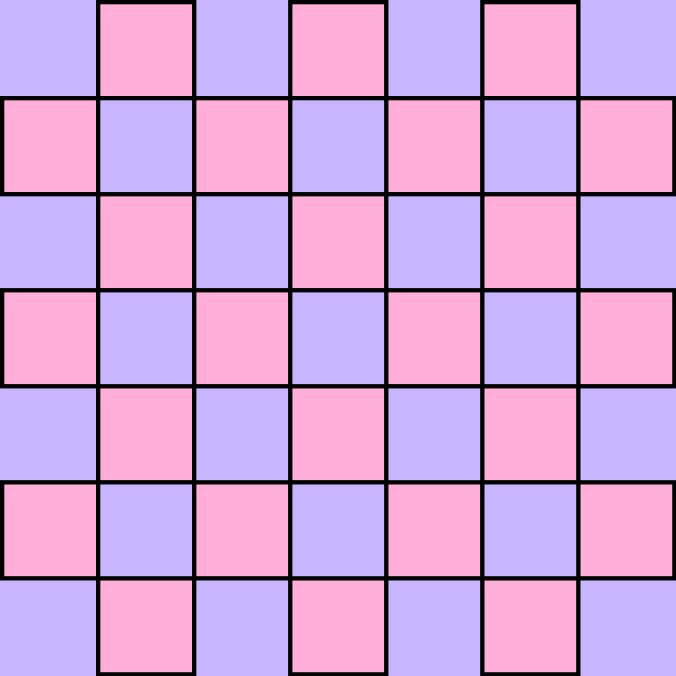
365 Day Forecast





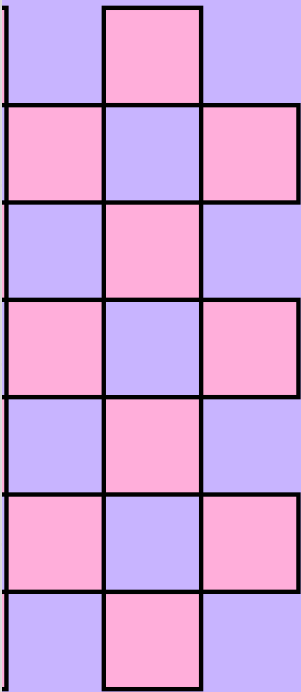
ENCODER-DECODER LSTM MODEL (SEQ2SEQ)

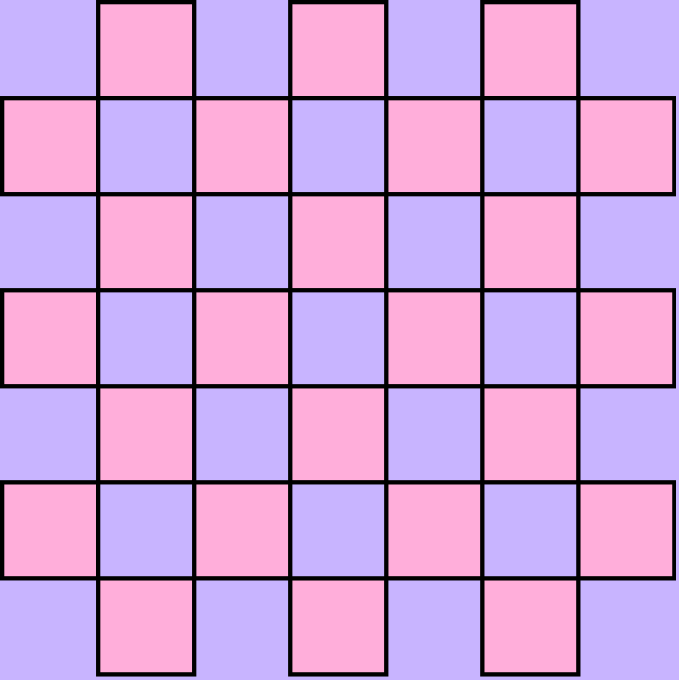




ENCODER-DECODER LSTM MODEL (SEQ2SEQ)

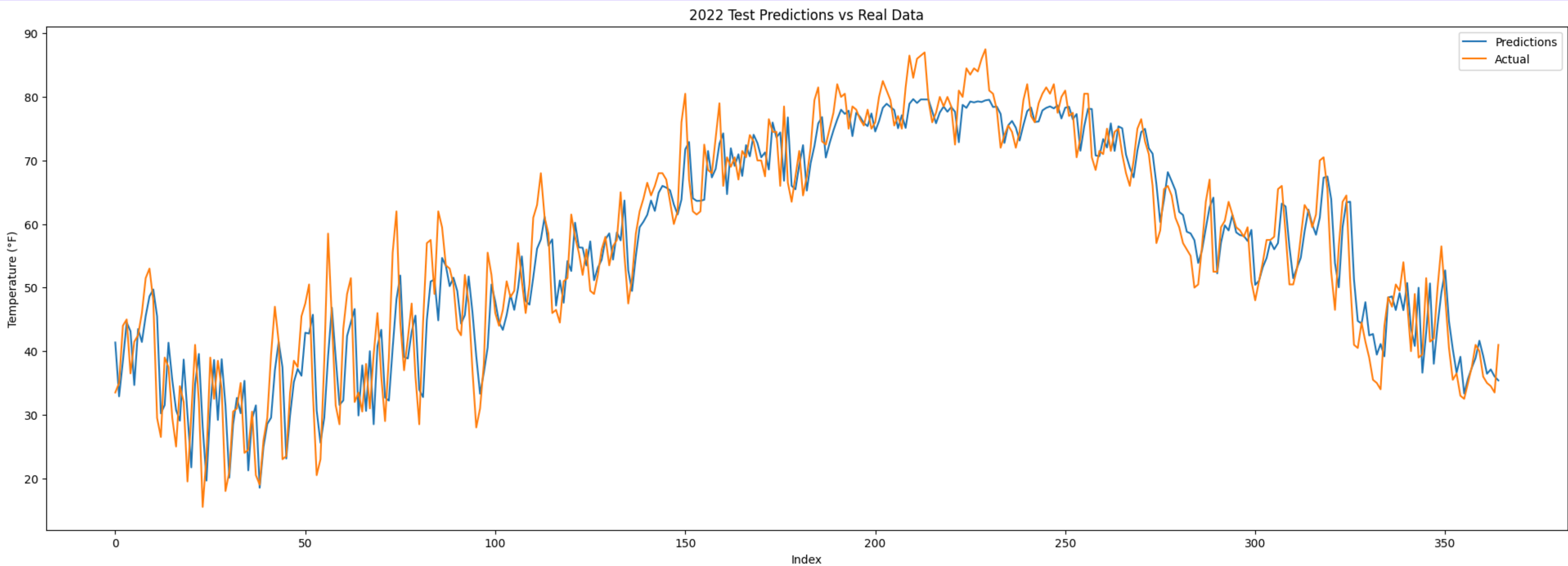
Model #	Epochs	Window Size	Horizon	Val loss	RMSE
1	10	30	60	64.1186	7.7996
2	10	30	7	57.2503	7.6941
3	10	7	7	61.0883	7.8908
4	10	30	15	66.831	8.2136
5	20	30	7	57.9485	7.6151
6	20	60	30	63.2914	7.7445
7	30	30	7	57.0479	7.5829





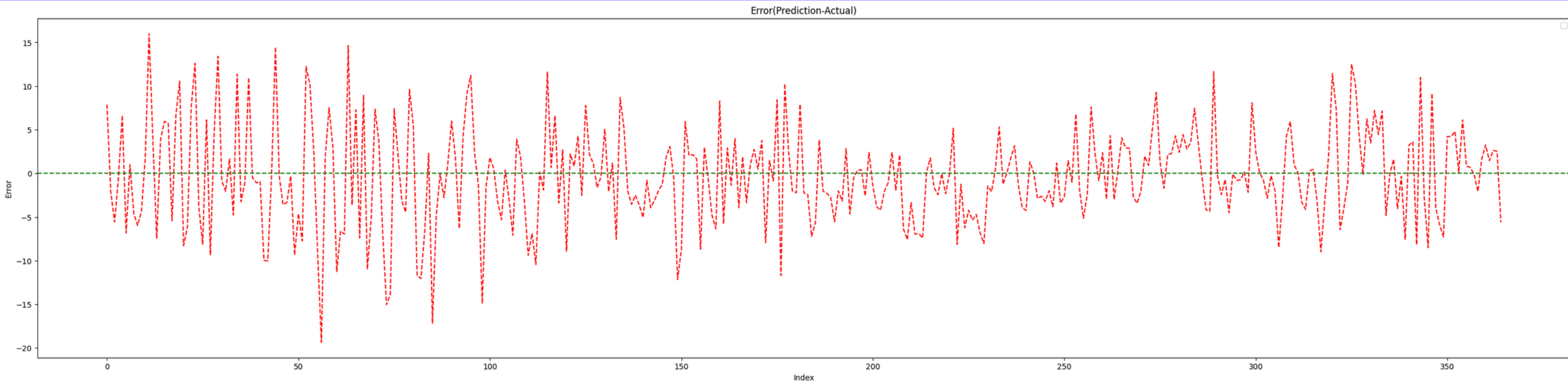
ENCODER-DECODER LSTM MODEL (SEQ2SEQ)

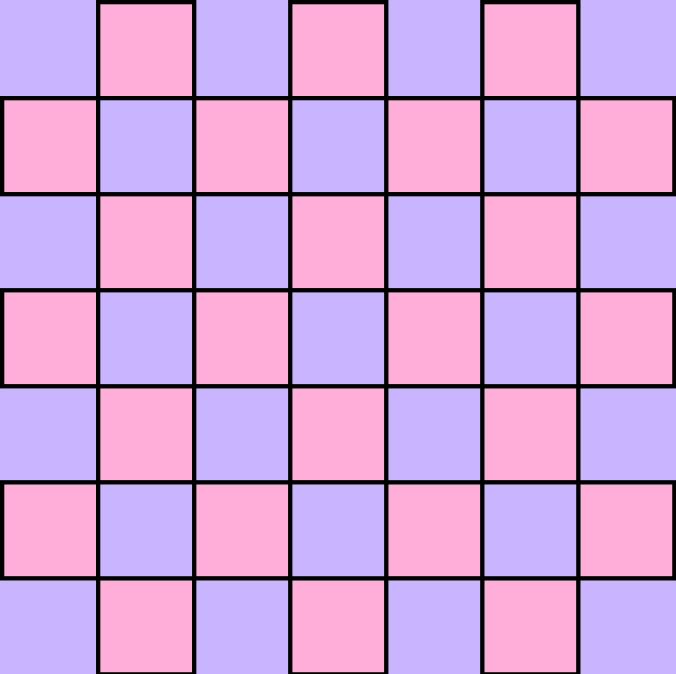
2022 Test Predictions vs. Real Data



ENCODER-DECODER LSTM MODEL (SEQ2SEQ)

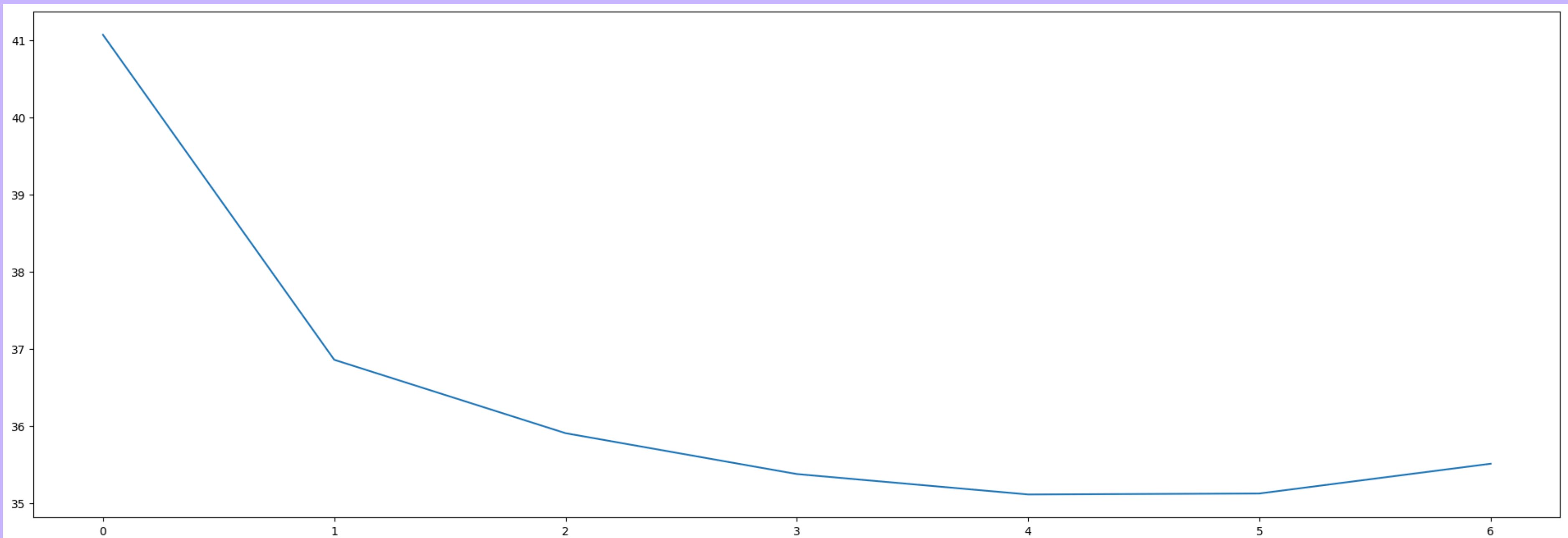
Error (Prediction-Actual)





ENCODER-DECODER LSTM MODEL (SEQ2SEQ)

7-Day Forecast



PROPHET

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

$y(t)$ = predicted value at time t

$g(t)$ = trend

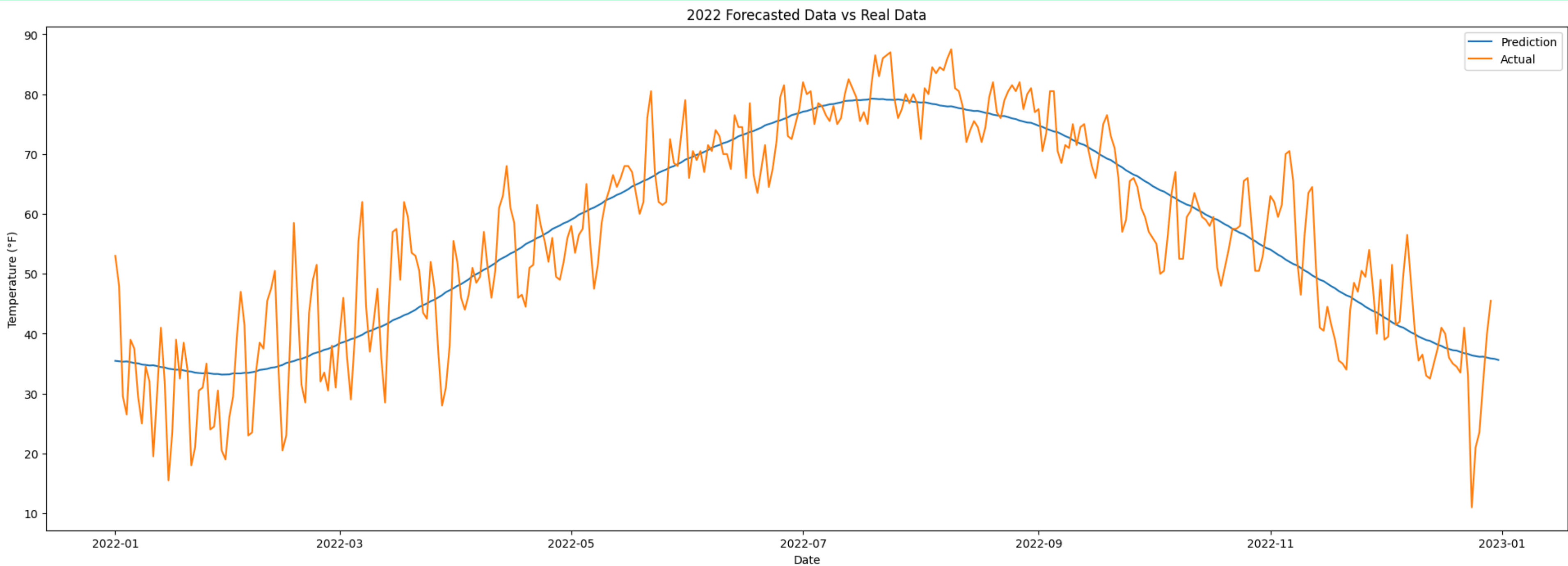
$s(t)$ = seasonality

$h(t)$ = holiday/event effects

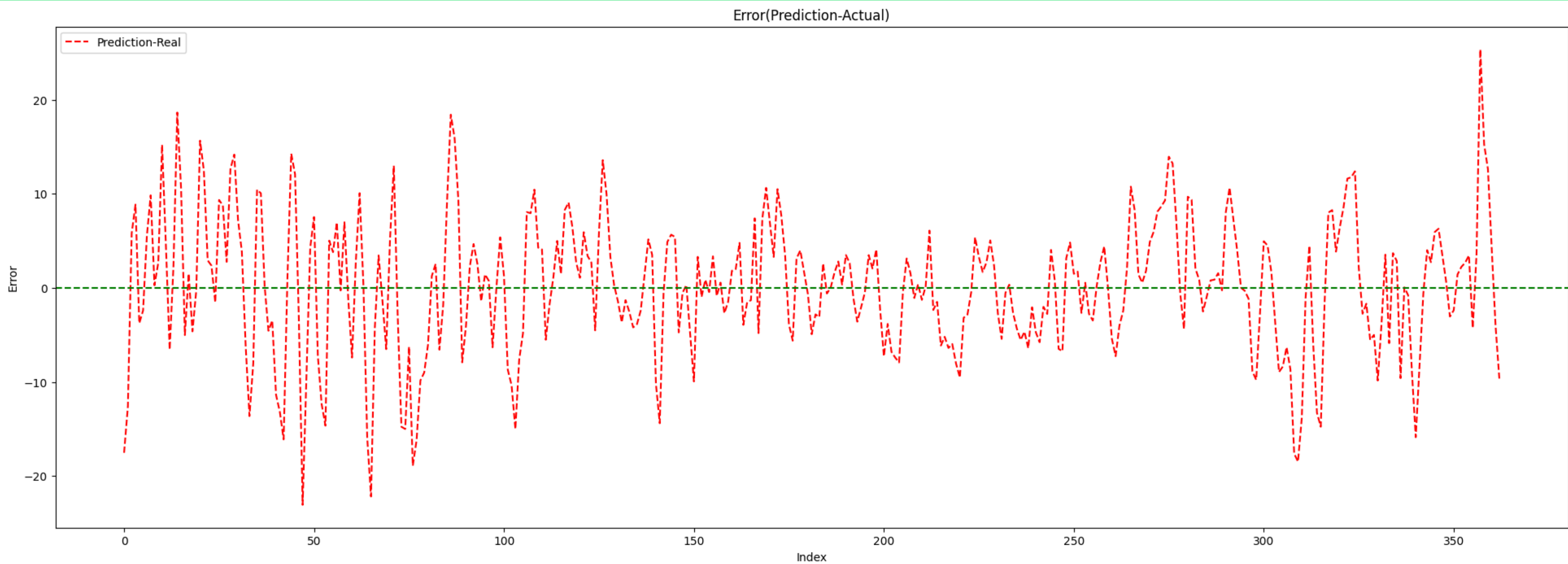
ε_t = error term

PROPHET

2022 Real Temperatures vs Forecasted Prophet Data

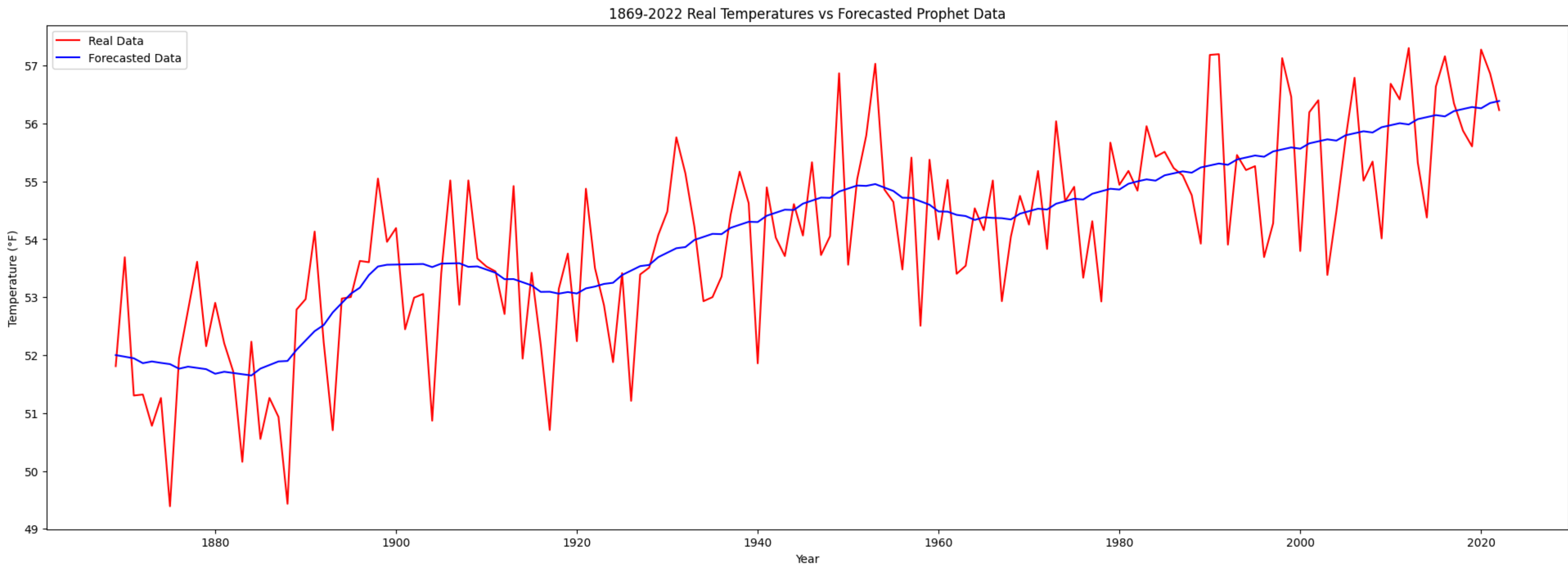


PROPHET

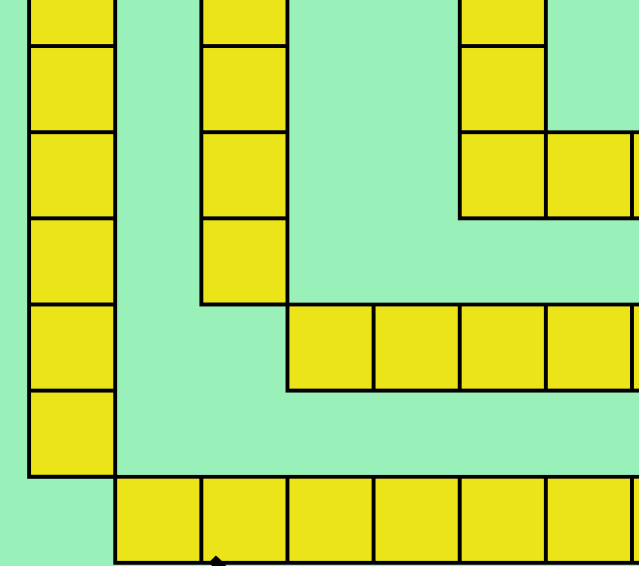


PROPHET

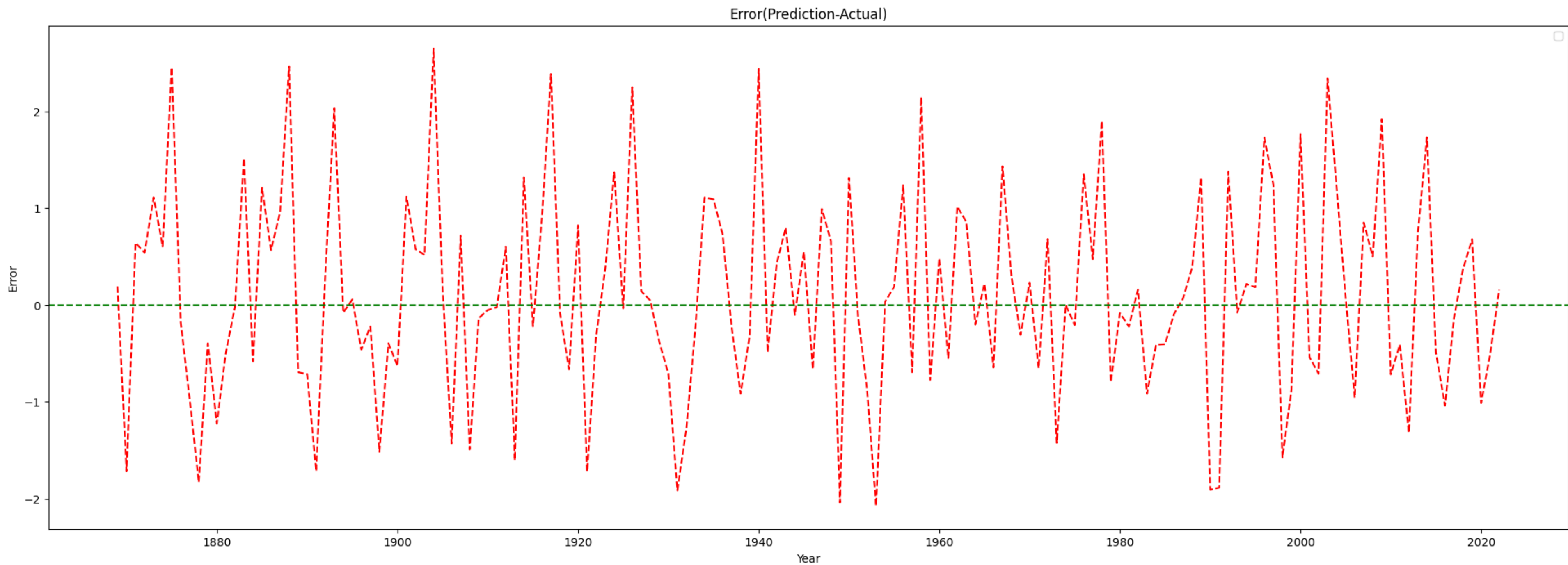
1869–2022 Real Temperatures vs Forecasted Prophet Data



PROPHET

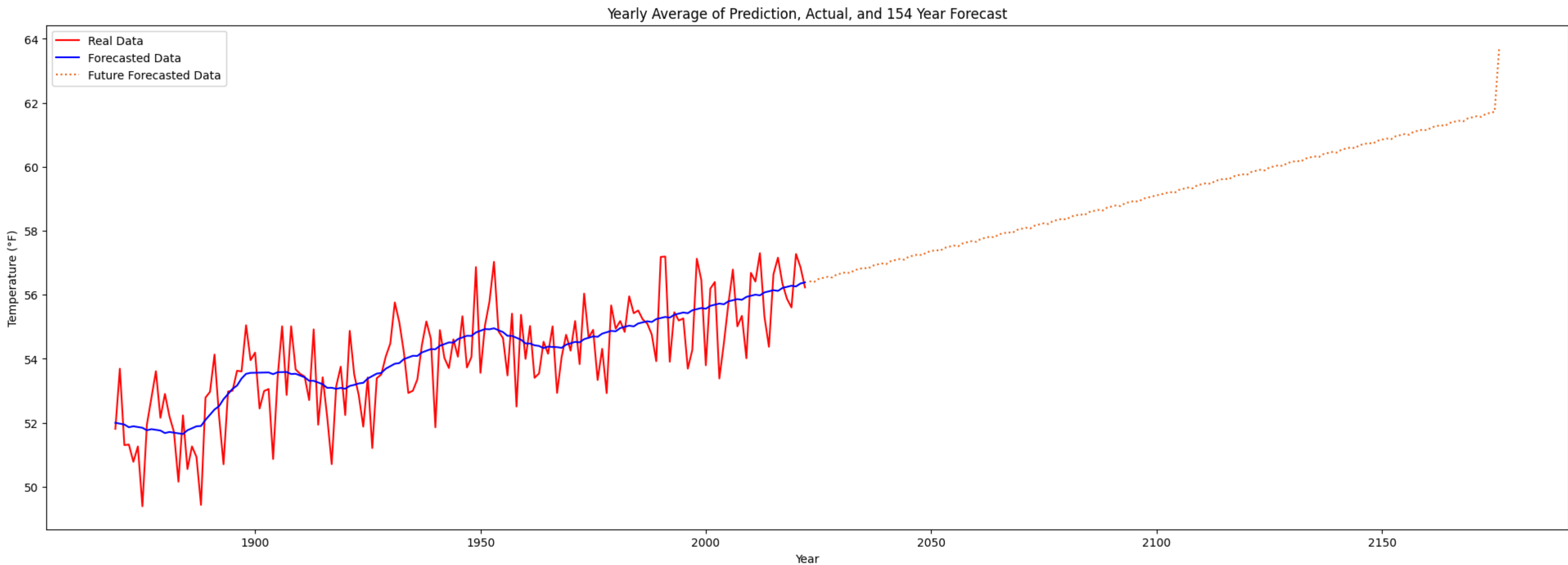


1869–2022 Yearly Average Error (Prediction–Actual)



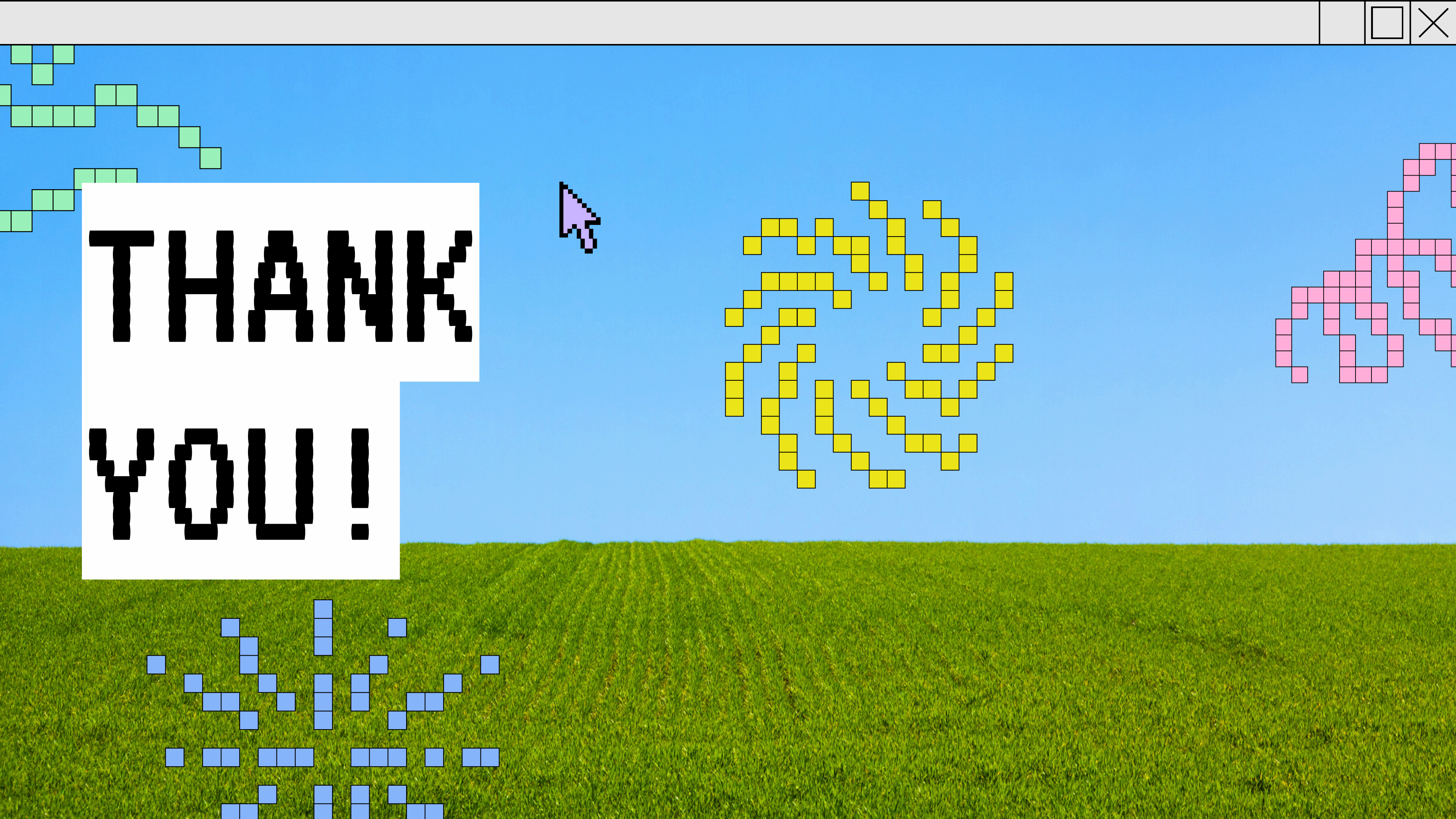
PROPHET

Yearly Average of Prediction, Actual, and 154 Year Forecast



MODEL COMPARISONS

Model	Mean Absolute Error
Many-to-One	4.08067696863987
Seq2Seq	4.21130120193312
Prophet	5.83632491030816



THANK
YOU!